

PERPETUAL FUTURE BLOCKCHAIN: A FRAMEWORK FOR MAPPING BLOCKCHAIN MARKET STRUCTURE

Author: Swen Werner

(8 July 2025)

The author is authorised to publish this article.

Abstract

This paper maps blockchain-native instruments and transactional structures to their closest financial counterparts – not to equate them, but to reveal how similar terminology can obscure fundamental differences in logic, function, and enforceability. It argues that using legacy financial terms such as *derivatives*, *settlement*, or *custody* to describe on-chain activity often leads to category errors. The paper proposes a structural framework for evaluating blockchain-based systems in their own terms, emphasizing determinism, message structure, and computational finality over legal form. It concludes that many crypto-financial instruments – especially perpetual futures and stablecoins – serve coordination or computational functions, not traditional financial purposes, and should be reinterpreted accordingly.

Keywords: Blockchain Market Structure, Tokenisation and Legal Frameworks, DeFi Protocol Architecture, Perpetual Derivatives, Smart Contracts, Financial Instruments, Stablecoins Value

THE MIRAGE OF FAMILIARITY

Regulators often start with **categories**, not **function**. They ask: “Which box do we fit this in?” Rather than: “What actually happens here – economically, legally, and operationally?”

- Blockchain **doesn't** represent a ledger— it's a *sequence of actions*.
- Tokens are not legal entitlements — they're *receipts or evidence* of an interaction.
- Without context (intent, identity, agreement), the “record” on-chain means nothing in legal terms.

A NEW FRAMEWORK IS NEEDED

We shouldn't ask:

- “Is Bitcoin a security?”
- “Is a smart contract a legal agreement?”

We should ask:

- What does this token **do**?
- Who created it, and what promises (explicit or implicit) attach?
- Can its meaning be understood outside of its technical record?
- Can it be enforced, settled, or reversed?
- How does the user know what rights they have?

This is where **legal engineering** is required — not shoehorning into legacy categories, but building new ones fit for purpose.

Unlike traditional systems which:

- Store **account states** (“Alice has €1,000”)
- Include **permissions** or authorisations (“Alice *may* transfer up to €500/day”)
- Reflect **rights and capacity**

...blockchain records only **executed transactions**:

“Alice sent 2 ETH to Bob at time T” That's it. There is no trace of what Alice *could have* done, or was *entitled* to do. And nothing can prove if Bob can control the 2 ETH other than by sending them when the transaction requires it. Any moment earlier is not proof.

NO PRIVATE DATA AND NO PRE-AUTHORISATION

By design, blockchains:

- Do **not** store off-chain identity
- Do **not** encode discretionary permissions
- Do **not** express forward-looking rights (like callable bonds, standing orders, authorisations)

They are optimized for:

- Public verifiability
- Minimal trust assumptions
- Deterministic execution of discrete events only now

Which **excludes**:

- Legal capacity
- Enforceable promises
- Forward-looking obligations

THE IMPLICATION FOR LAW & FINANCE

- **You can prove what happened** (past transfers, contract execution).
- **You cannot prove what could happen** (right to redeem, right to dividends, insolvency priorities).

That means there can be no legal rights without off-chain enforcement. Coherence between onchain and offchain status must be verified continuously or structurally guaranteed. Processes that have been verified to operate deterministically (without discretion of provider) can be considered equivalent to a legal duty

Why Most Tokenised Instruments Cannot Be Regulated as Securities. A legally valid security requires fixed rights, issuer accountability, and enforceability. Yet most smart contract-based tokens lack the structural and procedural integrity needed to meet those conditions.

NO ISSUER-VERIFIABLE LINK THERE IS NO BUILT-IN CRYPTOGRAPHIC OR LEGAL MECHANISM TO VERIFY:

- That the smart contract was deployed by the rightful issuer;
- That the contract represents a legally binding obligation.
- In traditional securities, the issuer is known, registered, and legally liable. On-chain, even if you see a contract, you cannot confirm who controls it or what it legally represents.

POST-DEPLOYMENT CONFIGURATION = MUTABLE RIGHTS

Any smart contract that:

- Allows whitelisting/blacklisting,
- Can change redemption logic,
- Adjusts transfer restrictions or roles,

...can retroactively alter the rights of the token holder. That directly violates certainty of entitlement, a cornerstone of securities law.

This makes such a system fundamentally unsuitable for defining ownership, legal title, or enforceable claims.

LACK OF DETERMINISTIC FINALITY A FINANCIAL INSTRUMENT MUST HAVE A FIXED MEANING AND FIXED OUTCOME ONCE CONTRACTUAL CONDITIONS ARE MET.

- If the rights can be disabled,
- The issuer can reconfigure the smart contract,
- Settlement relies on off-chain interpretation,

...then there is no legal certainty.

Put plainly: If I cannot prove today what I will own tomorrow, I don't own anything – I merely hold a revocable token.

SECURITIES REQUIRE IRREVOCABILITY TO FUNCTION AS A VALID BEARER OR REGISTERED INSTRUMENT, A TOKEN MUST:

- Bind the issuer to the rights described,
- Prevent post-issuance manipulation,
- Survive independent of issuer whim or network configuration.
- Any token that can be “updated” by admin keys or backend logic fails this test.

A REGULATORY-COMPLIANT DESIGN WOULD REQUIRE: IMMUTABLE SMART CONTRACT LOGIC FROM THE MOMENT OF ISSUE.

- Off-chain legal commitment clearly linking the token to contractual rights.
- Verifiable identity of issuer and control at all times.
- Legally binding governance of settlement and redemption.
- Without this – no token, regardless of naming, can be a valid “security”.

STRUCTURAL MISMATCHES WITH TRADFI

- Language ≠ Function

"Custody," "clearing," "settlement," and even "contract" are used by analogy, but the mechanisms differ.

A "token" is not a share, a "smart contract" is not a legal agreement, and an "oracle" is not a neutral information source.

- Always-On Finality ≠ Deferred Settlement

Every transaction is charged upfront (e.g., gas), often before it's fully resolved.

There is no concept of "netting" or DvP (Delivery vs. Payment) unless built explicitly — and even then, it's hard to verify synchrony.

- Clearing ≠ Risk Elimination

There's no default mutualization or central guarantee — it's point-to-point margining.

"Clearing" often means hope the code runs and the oracle doesn't fail.

- Oracles Are Not Passive

They are active intermediaries that introduce external dependencies and control — and thus must be regulated or trusted like agents or market data providers.

Technical Fragility & Operational Risk Is Structural Uptime is a mirage: No release management = bugs, forks, regressions, pauses.

Outages and fee spikes are normal (not exceptional). TradFi systems are fault-tolerant; DeFi systems often fail hard and fast.

"Permissionless" is operational fiction — someone controls contracts, upgrades, and oracle parameters.

Crypto-financial systems do not replicate the legal and economic safety of banks, brokers, or CSDs. They replace institutional trust with code, but in doing so they:

- Introduce new, opaque risk (oracle manipulation, admin key abuse)
- Offer no real-time regulatory visibility
- Lack legal recourse, unless paired with off-chain agreements

This should not be harmonized with TradFi regulation without first addressing these structural and legal mismatches.

BLOCKCHAIN ISN'T TRUSTLESS – IT'S TRUST-MINIMIZING

If a blockchain system is designed correctly, we shouldn't need legal enforcement, manual intervention, or off-chain trust. That's the ideal of a **self-executing, deterministic financial architecture**.

- **Every financial function** – custody, issuance, settlement – can be tokenized.
- If properly designed, **the logic of ownership and transfer is embedded** in the structure.
- **Users become nodes** in a peer-to-peer economic network with no need for third-party intervention.

However, even the most robust cryptographic system **cannot resolve a dispute about intent**.

FAT FINGER RISK IN DETERMINISTIC, COUNTERPARTY-MATCHED SMART CONTRACTS

Fat finger errors – accidental mispricing or execution – are structurally **impossible** in properly designed, deterministic matching environments. This is because they would require one of two conditions:

1. **Both parties to simultaneously and inversely misprice**, and still correctly match the counterparty – a near-impossible coordination failure.
2. **Faulty or manipulated smart contract logic**, which would invalidate the claim to determinism altogether.

As long as the network remains functional and the logic is auditable, there can be **no invalid transaction**, and thus, **no loss** from “accidental” trades.

ON-CHAIN MATCHING VS. OFF-CHAIN ORCHESTRATION

Today, DeFi platforms rely on **off-chain orchestration**:

- The user interacts with a front-end.
- The DeFi operator generates a signing payload.
- The user signs, and this triggers the smart contract to process the transaction.
- The protocol issues a **wrapped token**, through which rights to the asset are managed.

This structure begins with **centralised instruction management**, even if execution occurs on-chain. But if this process were reversed – with **the value-seeker issuing a zero-value reference token** (a request for risk) – and the counterparty responding with asset transfer upon receiving that reference – we would have:

- **Trust-minimised matching**
- **Blockchain-conformant booking**
- **Decentralised offer/acceptance flow**
- **Deterministic counterparty resolution**

SMART CONTRACTS AS MARKET INFRASTRUCTURE

In such a system:

- Tokens are not representations of legal rights, but **atomic instructions**.
- Matching becomes the event of legal significance – not the signing process.
- Value is not "sent", but **provoked** by a validated, structured request.

In traditional trading:

- A transaction is valid if **both parties agree**, even if one made a mistake (unless fraud is involved).
- The legal system **enforces the trade**, even against self-interest.

Smart contracts follow the same principle:

- If a transaction executes, it is binding – **unless** it violates pre-agreed logic.
- The logic is **auditable**, execution is **deterministic**, and accidental execution is virtually **impossible**.

Hence, smart contracts provide:

- **Automatic validation**
- **Minimal execution risk**
- **No post-trade ambiguity**

This **reverses** the current paradigm:

Traditional Model	Smart Contract Model
Sender initiates with assets	Receiver initiates with request
Execution relies on trust/intermediaries	Execution relies on deterministic code
Legal rights first, execution second	Matching logic first, legal implication second

REGULATORY IMPLICATIONS

- **Tokens are not assets**, but structured **messages**.
- **Ownership is not legal**, but **structural** – defined by the system logic.
- **Finality** is not relational or institutional, but **procedural** – validated on-chain.

This aligns more closely with **computational law** than with securities law.

KEY LEGAL PROBLEMS FOR TOKENS

However, this paradigm is **not yet recognised** in regulatory doctrine. Instead, regulators:

- Attempt to **force-fit crypto instruments** into legacy legal categories.
- Focus on **function over form**, without structural coherence.
- Ask "*What is this token?*" rather than "*What system structure defines its meaning?*"

Until the market structure changes:

- Enforceability of rights in the event of bugs, forks, or validator failure is unclear.
- Legal standing requires **form**, not just **function**.
- Without identity, courts cannot assign liability or uphold claims.
- **Finality** remains ambiguous without external reference points.

If a fat-finger trade occurs, or a validator disappears, or a protocol upgrade reassigns value:

- **Who decides what's correct?**
- **Who has standing to challenge it?**
- **Where is finality found?**

These are questions of legal ontology – not technical infrastructure – and they remain unanswered.

If we assume that *a blockchain record represents a legal right*, we

- Confuse *proof of action* with *proof of entitlement*.
- Create systems where bad actors (who can act) have more legal power than rightful owners (who may be unable to act).
- Undermine centuries of legal development built on *potential* capacity and *intention*, not just *execution*.

Tokens **do not represent** legal ownership **by themselves**. They are **logical operators** within a *specific system*. The meaning of a token is:

- **Structural** – defined by the system architecture (e.g., smart contract logic, API parameters, governance rules)
- **Relational** – it only has value in reference to other participants or protocols
- **Externalised** – meaning is not embedded in the token, but in its **interpretation** by systems or actors outside the chain
- **Ephemeral** – they do not exist as persistent data objects, but only as the logical consequence of a validated transaction between two or more nodes. Their existence is inferred from system state, not derived from any intrinsic identity.

A TOKEN IS NOT AN OBJECT

A token is meaningful **only if** multiple nodes reach consensus on a state update. It is the *effect* of a valid transaction applied to the prior state — not a self-contained asset.

State-based abstraction:

- Most tokens (e.g., ERC-20) are implemented as **account balances in smart contracts** — effectively a spreadsheet.
- The “token” is not a transferable object, but an **entry that updates** based on logic execution.

Proof of state, not asset:

- Holding a token is proof of a **certain network-approved balance**, not of a discrete, ownable object.
- This makes them unlike physical bearer instruments, which exist independently of consensus.

Tokens are not digital objects, but emergent records — state abstractions derived from consensus, not possession.

If someone takes over 51% of Ethereum, **they don't own it**, they just momentarily dominate its consensus. But:

- **Ethereum's value isn't in the code or the chain**, it's in the shared agreement on what it *means*.
- The community, developers, exchanges, and users **can just reject** the hostile version and continue on a fork.
- The original name, ticker symbol, and recognition will stay with the *social consensus*, not the attacker.

So:

Control of blocks ≠ control of value.

This makes 51% “attacks” more like **fork-triggering events** — not destruction. They're rejected by the very logic of decentralisation:

- Power without legitimacy fails.
- You can't force people to treat your fork as Ethereum if it breaks the rules they care about.

This is why blockchain is structurally different from banking, securities, or centralized infrastructure. It's **self-forking, permissionless infrastructure**. You don't *take over* Ethereum or Bitcoin — you *leave it*, and others decide whether to follow.

The law must be prepared to address scenarios in which a usurper chain—one not initially regarded as canonical—becomes the authentic source of truth through de facto validation. This is structurally analogous to an unconstitutional seizure of state power which, over time and through broad recognition, achieves legitimacy in the international legal order (Völkerrecht). The shift is not legal in origin, but political in effect, and the law subsequently adapts to the new reality.

TOKENS ARE NOT ASSETS

They are claims within a structure. In blockchain, value is not tied to any single token, but to the *structure* that the token signifies – and that structure can be recreated or replaced. A token only has value insofar as the structure around it – smart contract, protocol rules, network recognition – sustains and interprets it. If that structure is duplicated, forked, or superseded, the original token can become meaningless.

- A token from a smart contract that gets deprecated (or forked) is not “lost” – its meaning is lost.
- You can have 1 AAVE token from an abandoned contract, but it no longer entitles you to anything.

REISSUANCE OR FORKING ≠ INFLATION

This is very different from fiat systems:

- In traditional finance, creating a second “version” of an asset dilutes or counterfeits it.
- In blockchain, you’re not duplicating the asset – you’re replicating the structure. If people switch recognition to the new structure, the old token becomes the one with no value.

This undermines assumptions in law and finance where the asset is legally distinct and not structurally replaceable. In crypto, assets are not exclusive, and value is always conditional on social consensus.

OWNERSHIP WITHOUT RECOGNITION = EMPTINESS

Even if a token is “yours” (you control the private key), if the structure it’s part of is abandoned or forked, your ownership is economically void. This is why legal systems that focus on title or possession fail to fully capture the logic of crypto:

- Possession is not proof of entitlement.
- Recognition is not based on law, but on network choice.

Blockchain tokens are like pointers to meaning, not the meaning itself. You can own the pointer, but meaning can move – and ownership without meaning is worthless. Bitcoin is not defined by code or chain ID. It is defined by recognition. You can copy Bitcoin’s codebase and even its entire history – and still not have *Bitcoin*. Why? Because Bitcoin = a socially recognized rule set plus a consensus about its protocol.

- You can **be Bitcoin** in *code* or *structure*,
- But you’re **not Bitcoin** without *recognition*.

That’s why forks like Bitcoin Cash, Bitcoin SV, etc., are **technically close, but economically and socially distinct – they lack the *recognition layer* that sustains Bitcoin’s value**. This also proves that: In blockchain, identity is not technical – it’s normative. It’s a *shared interpretation*, not an objective property. You can’t claim ownership of “Bitcoin” without getting others to accept your definition. This is entirely unlike a centralized database or legal entity.

FINANCIAL INSTRUMENTS SURVIVE PROTOCOLS

Protocols are not financial instruments:

1. **A true financial instrument signifies value** – it either represents a claim, obligation, or right under private law (e.g., a bond, a share, or a swap). That right **persists** even if the infrastructure supporting it fails.
2. **Bitcoin and Ethereum do not signify legal rights**. They are **protocol-native artifacts** – they only exist *within* the blockchain context. If the protocol stops, the asset vanishes. There is no off-chain representation, no claim, and no fallback. That makes them **closer to synthetic exposures**, like perpetual futures – structurally incomplete and not “instruments” in any legal sense.
3. **Tokens that represent rights can survive the chain**. If a token maps to a legal obligation (e.g., a redeemable stablecoin backed by a contract or custody agreement), then the token is merely the *medium* – the legal value exists elsewhere. In that case, even if the protocol fails, **the right persists**, and recovery routes exist (e.g., legal redress, claims on collateral, etc.).
4. Therefore:
 - **BTC/ETH = synthetic exposure to an internal system** (like futures without expiry or delivery)
 - **Rights-bearing tokens = instruments linked to law**, capable of surviving even a protocol failure
 - **Conclusion: Value tied purely to protocol disappears with it. Value tied to law survives.**

ON-CHAIN FAILURE DOES NOT ERASE VALUE – IT DISPLACES IT

In decentralized systems, value is expressed through tokens and protocols – but **the economic logic doesn't vanish when the system fails**.

- A failed redemption contract still implies a **claim** (even if it can't be executed).
- A broken DAO vote may still **represent the will of holders**, even if the outcome isn't enforced.
- A collapsed bridge or DeFi protocol doesn't mean the **underlying obligation** is gone – only that the on-chain path to it is broken.

In legal terms, this means:

- Blockchain protocols **encode** performance but **do not constitute** legal rights.
- If a protocol fails, any value embedded in it **reverts to off-chain negotiation or legal recovery**.
- Therefore, **rights survive chain failure** – and we need criteria for **how to identify and assert them**, not deny their existence.

This mirrors the logic of **Howey**: the presence of an investment contract is not defined by form, but by economic reality. Similarly, the legal consequences of blockchain activity do not disappear just because the software broke.

CLARIFYING DEFI (AAVE) TRANSACTION FLOW: INTERFACE VS. LOGIC

When a user supplies assets to Aave via MetaMask, the interaction is not “direct” in any meaningful legal or architectural sense. Here's what actually happens:

1. **User visits Aave frontend** (typically app.aave.com) in a browser.
2. The frontend generates a **pre-formatted transaction** via JavaScript, using the Aave smart contract ABI.
3. This instruction is passed to **MetaMask**, which acts as:
 - A local signer
 - A UI for transaction confirmation
4. Upon signing, the transaction is sent to the Ethereum network for inclusion.
5. The Aave protocol smart contract, upon receiving the transaction, executes the function (e.g. deposit), updates internal accounting, and **mints a corresponding “aToken”** representing the user’s claim.

Key points:

- MetaMask does not verify that the transaction logic makes economic or legal sense. It just signs what’s presented.
- The frontend dictates what instruction you’re signing. It may be out of sync, manipulated, or spoofed.
- There is no native Aave identity or guarantee behind the aToken — it is minted by a contract, not issued by a person.
- “Secure” in this context means digitally signed, not legally protected.
- If a bug, front-end attack, or misconfiguration occurs, users have:
 - No counterparty to sue
 - No recovery method
 - No enforceable rights beyond what’s embedded in the code

This shows that what looks like a financial service is **actually a loosely-coupled flow between UI, wallet, and autonomous logic** — and any assumptions of regulatory equivalence are deeply misplaced.

A TOKEN ON A PERMISSIONED BLOCKCHAIN IS NOT DECENTRALIZED

This therefore inherits the regulatory obligations of traditional finance (TradFi) – not because of its digital format, but because of its centralized governance and control structure. 🔍

PERMISSIONED CHAINS HAVE IDENTIFIABLE CONTROLLERS A PERMISSIONED BLOCKCHAIN HAS KNOWN VALIDATORS, OPERATORS, OR ADMINISTRATORS WHO CAN:

- Control access
- Reverse or freeze transactions
- Define upgrade rules
- Enforce KYC or compliance policies

This reintroduces intermediary risk and regulatory responsibility – exactly what traditional systems already face.

FUNCTIONAL EQUIVALENCE TO EXISTING INFRASTRUCTURE A PERMISSIONED TOKEN SYSTEM IS FUNCTIONALLY SIMILAR TO:

- A centralized securities registry
- A clearing and settlement system
- A closed-loop payments network

NO TECHNOLOGICAL EXEMPTION BEING "ON BLOCKCHAIN" DOES NOT EXEMPT AN ASSET FROM SECURITIES, BANKING, OR PAYMENT RULES.

Regulators must look through the tech to assess:

- Who holds control
- Who can enforce rights
- Who bears responsibility for operations

If those functions are performed by a central party or consortium, the relevant financial laws apply – MiFID II, CSDR, PSD2, EMIR, etc.

Permissioned blockchains are not decentralized. They operate under identifiable governance and reproduce the core features of traditional financial systems. As such, any token issued on them must comply with existing financial regulation – because the risks and responsibilities are the same, regardless of technical implementation.

BLOCKCHAIN IS NOT ABOUT ENCRYPTION

It's about consensus. Blockchain is not "secure" because it encrypts data. It's secure because:

- Data is public (not private) and verified by every node.
- Security comes from **transparency**, **immutability**, and **redundancy**, not secrecy.

REDUNDANCY IS A FEATURE, NOT A FLAW

- Every full node keeps a **complete copy** of the blockchain.
- Every transaction is **re-executed** and **re-verified** by every node.
- This is *not efficient* in the computational sense – but **deeply resilient**.
- The more nodes, the **slower** the propagation and finality – but the **harder** it is to corrupt or compromise the network. It's the opposite of scalability engineering – instead of making things faster, it makes things harder to falsify.

ASYNCHRONOUS PROPAGATION IS KEY TO SECURITY

- Blockchain is **asynchronous by design** – there is no "master clock" or guaranteed order of receipt.
- Transactions propagate through a **peer-to-peer gossip protocol**.
- Finality is achieved only **after consensus**, not when an action is broadcast.

This means:

- **There is no trusted source of truth until consensus is reached.**
- **The system tolerates delay and disorder** in order to ensure **deterministic convergence**.

Blockchain flips the assumptions of traditional secure systems: it protects **not by hiding** data, but by making it **impossible to alter** without majority coordination. Security comes from **redundant, asynchronous transparency**, not centralized control or encryption.

Despite the theoretical symmetry implied by decentralised networks, **nodes are not equal** in practice. Each node maintains only a **partial view of the network**, informed by its **direct connections** – often a small subset of the total graph. This leads to:

- **Asymmetric propagation** of information
- **Varying latency** across geographic or infrastructural lines
- **Differentiated resilience** depending on local conditions

As a result, a node may **fall out of sync**, fail to validate a transaction in time, or miss a reorg – *through no fault of its own*. The network, however, is **unforgiving**: consensus proceeds without pause, regardless of a node's local experience.

DECENTRALISATION IS NOT AN EGALITARIAN GUARANTEE

This has direct implications for:

- Finality and fairness in transaction settlement
- Liability in the event of missed or invalid state transitions
- The limits of auditability for any single observer

Thus, decentralisation is not an egalitarian guarantee, but a structural fragility, where robustness is bought through redundancy – not uniformity.

BLOCKCHAIN DOESN'T SUPPORT TRADING

In traditional finance:

- **Trading:** You submit an *order* – a contingent offer to transact
- **Settlement:** You *perform* – cash and asset actually move

In crypto:

- Everything is immediate execution unless deliberately deferred.
- There is no abstract trading phase, unless built artificially (e.g., on DEXs via order books).
- Tokens don't represent claims, but *past activity* or *system states*.
- DvP is not native and requires smart contracts and thus no longer trust minimized
- In TradFi: Instruments define rights, and trades are promises. In DeFi: Tokens are *receipts of action but not necessarily instruments*, and transfers are *re-expressions of code*.

LIQUIDITY IS TIME-ZONE DEPENDENT

- Most liquidity concentrates during U.S. business hours, especially overlapping with Europe (e.g., 1–4pm GMT).
- Outside core hours, spreads widen, depth disappears, and execution risk increases – particularly for large or leveraged trades.
- This mirrors traditional FX or commodities, where technical markets may be open, but real liquidity is clustered.

NO "SETTLEMENT DAY" ON BLOCKCHAIN

- Blockchains operate on logical time but there's no absolute timestamp enforced by an authority
- Blocks just form a chronologically ordered chain. Finality is determined probabilistically (e.g., after n blocks, the data is "final enough")
- Blockchain's only notion of time is: "Did this happen before or after that?" – not "Did this happen at 3pm on Tuesday?" This destroys many core assumptions of traditional settlement systems: **No "close of business"** and no daily mark-to-market or cutoffs

ASSETS MUST BE REFERENCED *IN CHAIN-CONTEXT*

A digital asset can only be defined, understood, and regulated in reference to the chain or protocol it is natively tied to.

WHY?

1. No asset without chain context

- Bitcoin (BTC) on *Bitcoin* is fundamentally different from BTC on Ethereum (as wrapped BTC).
- A tokenized equity on a smart contract chain is not the same as the underlying equity.
- The asset's logic, security model, rights, and finality depend entirely on the native protocol and its rules.

2. Legal recognition depends on execution environment

- For an asset to have legal significance (ownership, transfer, enforceability), its **chain of origin** must be known.
- Off-chain representations (e.g. wrapped tokens) are **synthetic**, not the asset itself. They rely on trust or custodial intermediation.

3. Risk and valuation are chain-dependent

- Holding USDC on Ethereum ≠ holding it on Solana – risk profiles differ based on consensus, oracle systems, and smart contract logic.
- The *same token symbol* in a different execution context is **not the same financial instrument**.

4. Settlement, not symbol, defines ownership

- In traditional finance, ownership is anchored in **CSDs and legal registries**.
- In blockchain, **settlement occurs in the native ledger** – only that context defines actual possession or entitlement.

IMPLICATION FOR REGULATORS AND MARKET PARTICIPANTS:

Regulation must distinguish between:

- Native digital assets (e.g. ETH on Ethereum, BTC on Bitcoin)
- Synthetic or wrapped representations (e.g. wBTC, USDT on Tron)
- Contractual tokens representing off-chain claims (e.g. tokenized real estate or stock)

Otherwise, we lose legal clarity, fragment risk management, and introduce structural mispricing.

SMART CONTRACTS ≠ MARKET INFRASTRUCTURE: WHY UNIFORM SETTLEMENT IS ESSENTIAL

1. Smart contracts are siloed

Each smart contract – even on the same blockchain – operates in isolation. They cannot inherently share state, coordinate obligations, or guarantee global consistency. This leads to **fragmented liquidity, non-uniform settlement logic**, and an inability to construct coherent risk models across markets.

2. Securities issuance cannot be self-referential

A security cannot exist solely “within” a smart contract without violating core requirements of capital markets:

Without an interoperable legal and technical layer, the “security” is just a protocol artifact, not a legally binding financial instrument.

- Identifiable issuer and investor
- Rights and obligations traceable and enforceable
- Transfer and settlement that align with legal finality and market standards

1. Relay chains or asset-class specific settlement layers are necessary

To bridge siloed execution environments and coordinate global state across contracts, a **relay chain (or equivalent intermediation layer)** is essential. This allows:

- Shared price discovery
- Harmonized finality
- Uniform application of settlement rules per asset class

2. Settlement must not become the price signal

If every contract settles differently – by time, logic, or oracle input – then **settlement itself distorts the price**. That’s fatal for:

- Price transparency
- Arbitrage efficiency
- Market confidence

This is especially relevant for discussions about DeFi, stablecoins, or tokenized securities – and it clearly shows that **just putting financial assets “on chain” doesn’t create a market** unless the settlement architecture mirrors the coherence and consistency of traditional financial infrastructure.

INTERNET VS. BLOCKCHAIN: OPPOSITE INFRASTRUCTURAL LOGIC

While both are digital networks, **they're built for fundamentally different threat models** and operational assumptions.

Feature	Internet	Blockchain
Ownership	State-sponsored or corporate-owned (e.g., ISPs, IXPs, CDNs)	Globally distributed, run by private actors (miners, validators)
Access Control	Permissioned infrastructure; ISPs can gate access	Permissionless: anyone can run a node or deploy a smart contract
Trust Anchor	Domain Name System (DNS), IP routing, governed by ICANN and national regulators	Consensus algorithm and protocol code, governed by cryptography and open-source governance
Service Orientation	Designed to deliver content and services	Designed to record and enforce state transitions (i.e., token balances, contract logic)
State Role	States build, subsidize, or regulate infrastructure (e.g., state-owned telcos)	States are <i>adversaries</i> in many models (e.g., Bitcoin, Ethereum)

- The Internet depends on centralized trust in institutions to ensure continuity, performance, and security (IXPs, root name servers, fiber optic hubs).
- Blockchains assume those same institutions may be corrupt, untrustworthy, or subject to censorship – and therefore intentionally design infrastructure outside the control of states and corporations.

This is why crypto networks don't have IXPs – they *are* the exchange points. Every full node participates in validation and routing.

If the Internet is a highway system, built and maintained by public authorities for free movement of data, then the Blockchain is a wilderness trail – navigable by anyone, but each participant must carry their own provisions, map, and compass.

MARKET ABUSE PRESUPPOSES CONTROL OVER MARKET ACCESS.

Without control, there is no “market” in the legal sense – and thus no enforceable framework for abuse. In traditional finance, market abuse laws (like MAR, SEC Rule 10b-5, etc.) rely on the existence of:

- Designated market venues with admission criteria (e.g. LSE, Nasdaq),
- Gatekeeping intermediaries (brokers, clearing firms),
- Rules for transparency, order flow, and surveillance,
- Participants identified and governed by access rights.

Access is mediated → abuse is possible → legal framework applies.

BUT IN CRYPTO (DEFI/DEXES):

- No access control = **anyone can deploy, trade, or signal.**
- No listing process = **no gatekeeper or formal admission.**
- No legal venue = **no market operator with responsibilities.**
- No counterparty ID = **no enforceable disclosure duties.**
- Smart contracts have **no ability to screen participants.**

If the venue is open to everyone, and nobody has a duty to oversee conduct, there is no “market abuse” regime to apply.

If the regulator insists on treating DeFi or token DEXs as regulated markets, they would have to impose access control, identity, and admission structures – which contradict the design of these systems.

“If I can’t control market access, there can be no market abuse.”

“CONTROL” IS A RELATIONAL CONCEPT IN LAW BUT A TECHNICAL CONCEPT IN BLOCKCHAIN.

The regulation misuses the term ‘control’ to refer to blockchain private key possession, implying that technical ability equals legal custody. This is false. In blockchain, a private key does not establish legal agency, intent, or even identity. By treating mere technical access as ‘safeguarding,’ the regulation permits arbitrary transfers without enforceable trust structures. This collapses the legal distinction between custodianship and unilateral appropriation, effectively deregulating custody altogether.”

When crypto wallet provider say that they do not own or control the underlying software protocols which govern the operation of Digital Assets, this means they cannot control the asset regardless of have a private key or not. To claim "custodial basis" but stipulating investors bear all risk of loss, even if something goes wrong on their side collapses the legal meaning of safekeeping. A custodian owes duties: to protect, to segregate, to not misuse, and to return the asset. But that is structurally impossible for crypto assets.

There is no such thing as "a Bitcoin" in the sense of a tangible or self-contained object. You cannot own 1 BTC independently of the network's transaction log, which defines what that "1 BTC" means.

Bitcoin is not a coin — it is a protocol-based consensus log but a system that does not meet the required conditions of a financial ledger. What a user "owns" is not an asset, but the ability to generate a valid transaction that the network will recognize and include in its distributed transaction history.

This amounts to a right in action, not a right in rem. It is purely procedural: the network enforces no claims to ownership, records no parties or rights, and offers no guarantee of continued existence. A valid network can arise or disappear at any moment, and any "holding" is only meaningful insofar as the consensus rules continue to be maintained.

Therefore, no legal right to the asset exists independently of the technical protocol, and no remedy can be granted that contradicts the network's internal logic without destroying the asset itself.

"Transferable" requires that I can issue an order to transfer ownership. Under English law (and generally in financial regulation), transferability means:

- I can initiate a transfer (e.g., via instruction, trade, or contract),
- The system will reliably execute that transfer on my behalf,
- And a third party (e.g., a court, registrar, or clearinghouse) can verify or enforce it.

But if the issuer says: "We do not control the blockchain," then no one can guarantee transfer. If I cannot issue a transfer order to the issuer (like I would to my bank or broker), then:

- I have no enforceable right to a transfer,
- I cannot demand that the token be moved, redeemed, or replaced,
- "Transfer" becomes a self-help technical act, not a legal mechanism.

So any statement that a cryptoasset is "transferable" under the law is false or meaningless unless:

1. Someone legally obligated to act on transfer instructions exists,
2. That person can actually control the smart contract or asset, and
3. The transfer can be enforced under law.

Therefore: "Transferability" without legal control is a fiction. When Crypto Wallet Provider say "we don't control the protocol," they admit that: They cannot guarantee a transfer, They cannot enforce redemption, They are not acting like custodians or account managers. Thus, the asset is not legally transferable in the regulatory sense — only technically movable, if the user has access to a private key (which might not prove ownership anyway).

Any regulatory assumption that blockchain is a neutral or purely technological layer misrepresents its function. Blockchain rewires the architecture of financial instruments – replacing contractual interpretation with executable code, identity with pseudonymity, and legal discretion with automated logic. This is not an upgrade to traditional systems but a structural transformation. As such, blockchain-based financial instruments cannot be treated as equivalent simply by mirroring the *form* of traditional securities. The failure to grasp this distinction introduces serious legal risk and undermines the coherence of the existing framework.

BITCOIN'S DESIGN WAS FOR A SPECIFIC SCENARIO

The **Nakamoto whitepaper** explicitly describes a system for **peer-to-peer electronic cash**, where:

- Two **identifiable** people (physically or contextually) interact – e.g., *you stand in a shop*.
- The **payer knows the payee's address** and sends a fixed amount.
- There's **no need for future proof of ownership**, *because the transaction is complete the moment it's broadcast*.

That use case works without identity on-chain, **because identity is resolved off-chain at the point of transaction**.

When the same **transactional mechanism** is used for:

- **Securities** issuance (bonds, equity)
- **Custody** (long-term safekeeping)
- **Proof of ownership** (legal title or claim)
- **Redemption** or **legal rights** (like bearer bonds)

...it **collapses completely**, because:

- There is no on-chain **proof of identity or authority**.
- There is no way to **demonstrate control** after-the-fact.
- You cannot **interpret the meaning** of a token without understanding *who sent it, why, and under what terms*.

Every meaningful legal act (e.g., issuing a bond, transferring custody) must include:

- A **known actor** with legal standing.
- An **intentional declaration**.
- A way to **interpret and verify** what happened.

Bitcoin only solves the **finality of payment**, *not legal proof of entitlement*.

PERPETUAL FUTURES IN CRYPTO

A derivative under MiFID II (e.g. option, future, swap):

- Is tied to a maturity or expiry, enabling risk measurement over time.
- Is standardized, allowing clearing and transparency.
- Has a defined underlying and margin regime, enabling systemic oversight.

A perpetual swap:

- Has no expiry, no maturity ladder.
- Relies on a funding rate mechanism instead of settlement.
- Is often non-cleared, crypto-margined, and off-exchange.
- Creates open-ended exposure with indeterminate risk duration.

PERPETUAL FUTURES ARE STRUCTURALLY UNFIT FOR REGULATORY DERIVATIVE CLASSIFICATIONS

- Misclassification risk: Regulating perpetual futures as if they were traditional derivatives creates *false assumptions* about risk, margin, portability, and legal enforceability – a problem for market stability, investor protection, and systemic risk models.
- No true position: Unlike standard futures or options:
 - Perpetuals don't secure price advantage.
 - They reset exposure frequently, often every 8 hours.
 - You can't "carry" a strategic position without continual re-pricing.
- Lack of hedgeability: They *cannot be used* in a portfolio for hedging or capital efficiency – no fixed expiry, no guaranteed liquidity alignment with underlying assets.
- Rent extraction mechanisms: Funding rates redistribute capital to liquidity providers or market makers in a *non-transparent, non-prudentially regulated* environment. It's rent-seeking wrapped in a product.
- Overcollateralization + under-hedging: Traders must overcollateralize for volatility without gaining offsetting capital efficiency, leading to more systemic exposure and less control.
- The *funding rate* (the interest one party pays another to keep the position open) is **dynamic**.
- It changes every few hours and is often set by **market imbalances**, not by risk-neutral pricing models.
- It can **go negative**, flip directions, or spike dramatically during volatility. There's *no stable return* to project or hedge against. You can't price the instrument over a 24-hour horizon, let alone longer.

In case you hear that story: Robert Shiller did not "invent" perpetual futures in 1993. He built a framework for macro risk hedging, not this.

LACK OF TENOR LEADS TO UNHEDGEABLE RISK

Traditional derivatives have:

- A *defined expiry*, so risks and exposures are **bounded in time**.
- **Pricing models** (Black-Scholes, etc.) that rely on that fixed horizon.

But perpetuals:

- Have **no maturity**.
- Reset **every 8 hours**.
- Have floating funding that's uncapped and reactive.

No model can price risk over time because there is no time boundary. That means **hedging is impossible** beyond intraday trades. For most investors, especially non-professionals, **there is no coherent path to profit**. They are speculating on *direction*, while paying fees, volatile funding, and margin interest.

Perpetual swaps in DeFi are structurally incapable of supporting reliable pricing, hedging, or investment returns. The floating and reversible nature of positions, absence of expiry, and opaque, dynamic funding rates make them fundamentally **uninvestable** and **unpriceable** by traditional standards. Any attempt to regulate them as if they were MiFID-compliant derivatives introduces **systemic misunderstanding** of risk, undermines investor protection, and falsely equates volatility gambling with financial hedging.

MiFID II – Annex I, Section C These clauses define financial instruments under EU law. The relevant points are:

- Section C(4)–(10) covers *options, futures, swaps, forward rate agreements*, and other derivative contracts, provided they are settled at a future date, and in many cases on regulated markets.
- For example, C(4): "Options, futures, swaps, forward rate agreements and any other derivative contracts relating to securities, currencies, interest rates or yields, or other derivatives instruments, financial indices or financial measures which may be settled physically or in cash."

The entire structure implies a defined event – a settlement date, maturity, or payoff – that is quantifiable, hedgeable, and legally enforceable.

CRYPTO MARKETS CONTRADICT CORE FINANCIAL THEORY.

But That Doesn't Mean the Theory Is Wrong. We are wrong how we apply it,

What we observe in crypto directly contradicts some foundational principles of financial theory. Tether dominating despite high cost, an undifferentiated product. It doesn't guarantee redemption, and it's expensive to move on Ethereum. And yet, Ethereum 'leading' despite structural cost inefficiency

And any explanation cite is not more a cute story not an argument..

- It was **first** to support real DeFi and NFTs,
- It has the **largest developer ecosystem**,
- It is considered "**legitimate**" by institutions.

None of this means anything. Staking as "security" serves no purpose. It is mostly a simulation of responsibility, not actual capital risk and functionally similar to inducements And validators don't offer added value other than following EVM rules.

In crypto, the **real asset is the herd**, and **you can't take it with force**. Try, and they'll just **walk to another pasture**.

Ethereum-like systems, where each asset has bespoke logic (smart contracts), break the clean separation between trading and settlement. This distorts the price signal and introduces randomness that traditional financial infrastructure avoids.

But Stablecoin are somehow infrastructure-ish and we do not have Coca Cola Shares doing something like this for say a stock loan involving Apple Shares and helping out DTCC a bit. But that is exactly true for Stable Coin.

But why?

THE EVM IGNORES THE OFF-CHAIN COLLATERAL OF STABLECOIN – SO IT DOESN'T EXIST ON-CHAIN

- The Ethereum Virtual Machine (EVM) doesn't "see" or "verify" what backs a stablecoin.
- It doesn't know if 1 USDC equals \$1.
- It doesn't know if Circle is solvent.
- It doesn't know if tomorrow exists.

ON-CHAIN, THE EVM ONLY RECOGNIZES:

- Token contracts with internal logic (balances, transfers),
- Not external guarantees or legal contracts.

On-chain, “stable” is not a function of any of any off-chain collateral. Stablecoins claim to represent an external state (“\$1”). But that state is never validated by the system that moves the coin. Therefore, on-chain it has none

So even if someone says: “Don’t worry, USDC is backed 1:1 by Treasuries.” A DeFi protocol **cannot and should not** differentiate that from: “My cousin promises he’ll pay you back.” because the DeFi ability to exercise the right of redeeming on their respective premise is the same: none.

In decentralized systems:

- A “stablecoin” is **just a token with a price = 1 label**.
- That **label** is used as an **input** to protocol logic (e.g. collateral ratio, liquidity pool math).
- The actual **stability** is *not enforced* by anything **on-chain** unless there’s a **mechanism** (e.g. DAI’s liquidation engine or crypto overcollateralization).
- The reason is that we need this info in token form to enable the DeFi construct

FIAT-ED COINS IS THE ANSWER WHEN BACKED VS. UNBACKED IS IRRELEVANT ON-CHAIN

If I use USDT or a made-up SWisNr1 in a lending protocol, the smart contract doesn’t **verify backing** – it uses the **peg as a logic rule**.

- The **utility** of a stablecoin **within** DeFi comes from:
- Token liquidity,
- Protocol recognition (token.symbol == "USD"),
- User expectation of price = 1

Tether makes billions by arbitraging interest on reserves, but the users **don’t get value from the reserves**, they get value from the **symbolic function of 1 USDT = 1 USD inside the DeFi machine**.

And this function doesn't require collateral.

- If I mint an unbacked token that I promise will always be worth 1,
- And I make it usable in DeFi protocols,
- Then it's **functionally indistinguishable** from a "backed" coin like USDT. **Unless** you leave DeFi and go redeem it off-chain. But SWisNr1 is only a technical placeholder it has no value otherwise.

THE "BACKING" DOESN'T EVEN BENEFIT IF YOU EXIT TO FIAT. THE STABLE IN STABLECOIN IS A DATA ATTRIBUTE – NOT A PRICE STABILIZATION MECHANISM.

- Stablecoins don't exist because people want dollars. They exist because DeFi needs a conversion convention in the form of token – and USD via USDT just happens to fill that slot. But the collateral is irrelevant, the value can be declared even if it has none. Their function is computational, not economic.
- **Liquidity pools reveal this structure**

Then: $\text{Apple} / \text{Mercedes} = (\text{Apple} / 1\text{USD}) \times (1\text{USD} / \text{Mercedes})$ The stablecoin cancels out. It is the **pivot**, the **intermediate constant**, not the economic object.

- A pool of Apple ↔ 1USD
- A pool of Mercedes ↔ 1USD
- The question is not whether a stablecoin can be redeemed at par, but whether its "1" can be used as a computational anchor. That's why algorithmic stablecoins function – not because they are solvent, but because they are syntactically coherent as long as we use them correctly. They resolve cross-asset relationships through a shared unit, not a shared store of value.
- DeFi doesn't need "backing" – it needs predictable behavior.
- The so called peg is a label but works without backing. We don't need to give a company 100 billion to remind us of '1 USD'. That is the value proposition of Tether but there is no need for collateral.
- An unbacked token that simply allows to technically express a relation against a dollar is equally **useful** inside a smart contract if we use it only for this purpose. Stablecoin are a confusion. Backed or unbacked makes no difference. DeFi has no stablecoins, only a technical reference to 1.

In traditional finance, stability implies solvency and backing.

In DeFi, stability implies **consensus on expected behavior**, not collateral.

This decouples the label from economic substance and reframes stablecoins as **coordination primitives**, not financial products.

BLOCKCHAIN AND LAW: A STRUCTURAL ISOMORPHISM

Blockchain is not just a technology — it is a structural representation of law itself. The relationship is *isomorphic*: the components of a functioning blockchain system map one-to-one with the foundational logic of legal order under conditions of freedom.

- Nodes = Legal Subjects (Persons): Each node acts as a sovereign agent. It validates, accepts, or rejects changes to its state. Like individuals in a legal order, it retains autonomy but must respect the shared rules to participate.
- EVM / Consensus Logic = Arbiter of Coherence: The Ethereum Virtual Machine (or equivalent consensus logic in other chains) enforces rule-consistency — not based on morality, but on structure. It acts as a structural judge ensuring that all operations follow the shared constraints of the protocol.
- Smart Contracts = Legal Code: Like statutory laws, smart contracts encode permissible actions and consequences. They are deterministic, auditable, and universally applied across the system.
- Protocol = Constitution: The protocol defines the constitutional foundation. It may be updated (via fork or governance) — just like a human legal order can amend its constitution — but only through structural consensus.
- Mining / Staking = Compromise Mechanisms: These are historical artifacts designed to maintain synchrony under economic incentive — but they are not essential to the legal isomorphism. If state progression were made deterministic (e.g. FIFO + loop-breaker logic), you could have a fully law-like system without mining or staking.
- Forking = Constitutional Crisis or Revolution: When consensus breaks, the chain forks. This mirrors how nations break into parallel legal orders when foundational agreement collapses.

This framing reveals blockchain not as “code-as-law,” but law-as-code — a formal, operational instantiation of mutual non-interference, rights recognition, and structural governance. Native Tokens as Logical Property: Structural Necessity in Blockchain

As argued in *The Physics of Law* (Werner, 2025), private property is not a moral right, but a structural tool: it enables multiple agents to act freely within a shared space without obligating or interfering with one another. It is the logical partition that stabilizes absolute freedom under conditions of mutual non-constraint.

Blockchain systems implement this principle natively.

A native token, owned by each node, is not merely an asset — it is the *structural analogue of property*. It marks the boundary of agency. Just as human societies require property to prevent the actions of one individual from intruding upon another, blockchain nodes require tokens to define who can do what, with what stake, and at what boundary of control.

This aligns directly with the conditional axiom of lawful structure:

“I will not constrain your freedom, as long as you don’t constrain mine.”

Tokens, then, are not instruments of exchange — they are markers of **participatory legitimacy**. They denote capacity to act *without interference* and make possible the coordination of decentralized, sovereign agents.

Just as legal systems require property to function coherently without top-down command, **blockchains require native tokens** to partition control, allocate state transitions, and preserve order without central authority.

Without this logic, decentralization collapses into either chaos or covert centralization.

LAW AS COHERENCE, TOKEN AS CONFIRMATION

Law is not embedded in institutions — it is embedded in the **coherence between action and mutual recognition**. A lawful act is not one sanctioned by authority, but one whose **structure preserves non-constraint**, enabling actors to meet without interference.

In blockchain systems, the **token is not the asset** — it is the **logical confirmation** that such coherence has been achieved. It affirms that:

- A valid offer was made,
- A valid response was recorded,
- And the state transitioned deterministically without violating any constraints.

This means **the rule of law is not immanent to any blockchain**. It must be **tested**, verified, and recognized. A chain only *becomes* a lawful structure **when its processes consistently uphold structural justice** — i.e., non-interference, determinism, and symmetry of action.

Therefore, private and public law must prepare for a world in which *lawful status is demonstrated through structural logic*, not merely asserted by legal code.

RECOGNISING LEGAL STANDING THROUGH STRUCTURAL GOVERNANCE

In decentralised environments, traditional markers of legal status — such as identity, jurisdiction, or contractual formalities — are often absent. Yet, law does not disappear. Instead, **lawful governance arises from structural coherence**, not institutional identity. We propose a test of legal recognition grounded not in credentials, origin, or contractual formality, but in **how nodes behave** within a system.

A peer is recognised by their *lawfulness of governance* — their adherence to the structural principle:
“I will not constrain your freedom, as long as you don’t constrain mine.”

In blockchain systems, this principle manifests as **trust-minimised deterministic processing**. In law, we call it **the rule of law**.

This parallel reveals a deep isomorphism between decentralised consensus and lawful order: both systems define legitimacy not by authority, but by behaviour — specifically, by structural compliance with mutually agreed constraints.

This recognition is *functional*, not formal. It asks:

- Does this actor participate in a shared protocol that guarantees non-interference?
- Does their conduct enable state changes that are deterministic, auditable, and symmetric?
- Is the system structurally capable of identifying and correcting constraint?

If so, legal effect may attach not to the actor's identity, but to **the structural role they fulfill** within a lawful system.

This is essential if legal systems are to meaningfully engage with decentralised protocols, where:

- Authority is not assigned but **emerges** through process
- Agreements are validated **by structure**, not signature
- Finality arises **from coherence**, not institutional assertion

REFERENCE

Werner, Swen, THE PHYSICS OF LAW (July 06, 2025). Available at SSRN: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=5341278